

PSEUDOCODE: Longest Balanced Substring

Goal:

Find the longest substring that
contains exactly 2 distinct
characters
with equal frequencies.

Example balanced substrings:
"aabb", "abab", "xyxy"

FUNCTION isBalance(String,
start, end):

```
array frequency[26] = {0}  
different_chars = 0
```

```
FOR i FROM start TO end:  
    index = String[i] - 'a'
```

```
        IF frequency[index] == 0
THEN:
        different_chars =
different_chars + 1
        END IF
        frequency[index] =
frequency[index] + 1
    END FOR
```

```
IF different_chars != 2 THEN:
    RETURN false
END IF
```

```
value1 = 0
value2 = 0
```

```
FOR i FROM 0 TO 25:
    IF frequency[i] > 0 THEN:
        IF value1 == 0 THEN:
```

```
    value1 = frequency[i]  
ELSE:  
    value2 = frequency[i]  
BREAK
```

```
END IF
END IF
END FOR
```

```
RETURN (value1 == value2)
END FUNCTION
```

```
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```

```
FUNCTION startFunction(String,
i):
```

```
    IF i == Length(String) THEN:
        RETURN 0
    END IF
```

```
option1 = endFunction(String, i,
i)
```

```
    option2 = startFunction(String,  
i + 1)
```

```
    RETURN max(option1,  
option2)  
END FUNCTION
```

FUNCTION endFunction(String, i,
j):

IF $j == \text{Length}(\text{String})$ THEN:

RETURN 0

END IF

currentLength = 0

IF isBalance(String, i, j) THEN:

currentLength = $j - i + 1$

END IF

RETURN max(currentLength,
endFunction(String, i, j + 1))

END FUNCTION

FUNCTION

longestBalanced(String):

 RETURN startFunction(String,
0)

END FUNCTION

MAIN:

 INPUT String

 result =

longestBalanced(String)

 OUTPUT "Longest Balanced
Length: " + result

END MAIN